Statistics & Econometrics 2025

Tutorial 4: Problem Set

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1 Ordinary Least Squares Regression

Exercises to deepen understanding of OLS estimator for univariate linear regression and corresponding standard errors.

a) You have 12 years of time-series data on your firm's portfolio and want to compare its performance to a benchmark portfolio. Using these data, your intern has computed the sample annual average of your firm's portfolio return to equal 8% and that of the benchmark portfolio to equal 6%. (S)he further calculated the sample annual standard deviation of your firm's portfolio return was 5% and that of the benchmark 8%.

   Question: Is your manager right if she says the firm's portfolio performs better while being safer than the benchmark portfolio?

   Solution: Call the benchmark portfolio x and the firm's portfolio y. Then, compare the sample means and variances of y and x. If the sample mean $\bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t$ is greater than the sample mean $\bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$, then your firm's portfolio has delivered higher returns in the sample. This is the case for your firm's portfolio. Likewise, if the sample variance $Var(x) = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^2$ is greater than the sample variance $Var(y) = \frac{1}{T-1}\sum_{t=1}^{T}(y_t - \bar{y})^2$, then your firm's portfolio has been less risky than the benchmark over the sample considered. This is the case here as well for your firm's portfolio. Therefore, your manager's claim is correct.

b) Now your manager asks you to find out what happens to the firm's portfolio if the return of the benchmark portfolio changes by 1 %. In addition, (s)he is interested if your firm's portfolio earns a larger return than the benchmark portfolio, once we adjust for exposure to the benchmark portfolio. To help you in your task, your intern calculated that the sample correlation between the two portfolios equals 80%.

   Question: Calculate the β and α of your firm's portfolio against the benchmark.

   Solution: To estimate the quantities your manager is interested in, you run a linear regression of your firms' portfolio return on the benchmark portfolio return:

$$y_t = \alpha + \beta x_t + u_t.$$

   Estimating the regression using OLS, your estimates for α and β will be

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

and

$$\hat{\beta} = \frac{Cov(y,x)}{Var(x)}$$

where $\bar{x}$ and $\bar{y}$ denote the sample means of the two portfolios, $Var(x)$ the sample variance of the benchmark portfolio return and $Cov(y,x)$ the sample covariance between the two portfolio returns. You have been given information on the sample correlation and sample standard deviations for the two portfolio returns. So before you can use this estimator, you need to convert these into the sample variances and sample covariance. The definition of the variance is the square of the standard deviation. So a sample standard deviation of 5% (8%) translates into sample variances of 25 and 64, respectively. The correlation is defined as the covariance divided by the product of the standard deviations:

$$Corr(x,y) = \frac{Cov(x,y)}{\sigma_x \sigma_y}$$

where $\sigma_x = \sqrt{Var(x)}$ and $\sigma_y = \sqrt{Var(y)}$. Hence, you can impute the sample covariance as

$$Cov(x,y) = Corr(x,y) \cdot \sigma_x \cdot \sigma_y$$

Given the numbers above, this is

$$Cov(x,y) = 0.8 \cdot 8 \cdot 5 = 32$$

Now you have all the ingredients to compute the coefficients of the linear relationship between y and x. Plugging into the OLS formulae for $\hat{\alpha}$ and $\hat{\beta}$ you obtain

$$\hat{\beta} = \frac{Cov(y,x)}{Var(x)} = \frac{32}{64} = 0.5$$

and

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 8\% - 0.5 \cdot 6\% = 5\%$$

Hence, you tell your manager that the firm's portfolio has a beta of 0.5 and an alpha of 5%.

c) Your manager is surprised and says that for a correlation of 80% between the original portfolio and the benchmark portfolio, (s)he would have expected a beta equal to 0.8

Question: How can you test your manager's hypothesis? And can you reject your manager's hypothesis at the 5% level? For this analysis, assume that the estimated standard error for β_y was $SE(\hat{\beta}_y) = 0.125$.

Solution: Your manager's null hypothesis is $H_0 : \beta_y = 0.8$ vs. $H_1 : \beta_y \neq 0.8$

You can test your manager's hypothesis with the test of significance approach that you remember from your Statistics class. Luckily, you recall that the t-test statistic is given by

$$t_\beta = \frac{\hat{\beta}_y - \beta^*}{SE(\hat{\beta}_y)} \sim t_{T-2}$$

You have twelve years of data, so $T-2 = 10$. Moreover, β^* denotes the hypothesized true value of β which in this case equals $\beta^* = 0.8$. With these inputs, your test statistic is

$$t_\beta = \frac{\hat{\beta}_y - \beta^*}{SE(\hat{\beta}_y)} = \frac{0.5 - 0.8}{0.125} = -2.4$$

You have to compare this to the critical value for a two-sided t-test with $12-2 = 10$ degrees of freedom: $t_{crit} = t_{10;5\%} = 2.228$. Hence, since $t_\beta = -2.4 < -t_{crit} = -2.228$ you can reject your manager's (null) hypothesis at the 5% level.

To make your manager a little happier you note, however, that $t_{crit} = t_{10;1\%} = 3.169$. Hence, while you can reject his hypothesis at the 5% significance level, you cannot reject it at the 1% level.

d) Your manager is still somewhat skeptical. Show him/her that the confidence interval approach that you also learned about in your class gives rise to the same conclusion.

Specifically, you recall that the hypothesis can be tested by constructing the confidence interval for $\hat{\beta}_y$ and checking whether the hypothesized value of $\beta_y^* = 0.8$ falls inside this interval or not.

Solution: The confidence interval is given by $(\hat{\beta}_y - t_{crit} \cdot SE(\hat{\beta}_y); \hat{\beta}_y + t_{crit} \cdot SE(\hat{\beta}_y))$.

You remember that for a 95% confidence interval the critical value is $t_{crit} = t_{T-2;5\%}$.

Hence, it is the same as for a $t-$test with significance level 5%. You can therefore use the same critical values as before.

With these pieces of information your 95% confidence interval for $\hat{\beta}_y$ is

$$(0.5 - 2.228 \cdot 0.125; 0.5 + 2.228 \cdot 0.125) = (0.2215; 0.7785)$$

Since your manager's hypothesized true value $\beta_y^* = 0.8$ falls outside this interval, you can again reject his hypothesis based on a 95% confidence interval. You explain to him that this means that if you had observed a 100 different samples of portfolio returns, only in 5 cases would the hypothesized value of $\beta_y^* = 0.8$ fall into that interval.

That said, you also tell him that his hypothesis falls into the 99% confidence interval since $(\hat{\beta}_y - t_{10;1\%} \cdot SE(\hat{\beta}_y); \hat{\beta}_y + t_{10;1\%} \cdot SE(\hat{\beta}_y)) = (0.5 - 3.17 \cdot 0.125; 0.5 + 3.17 \cdot 0.125) = (0.104; 0.896)$ includes his hypothesized value of 0.8.

2 Restricted Regressions

a) Assume your colleague also shows you two other regressions that he ran but didn't include in the presentation. The first was a regression of the change of the unemployment rate on the change of the fed funds rate:

$$\begin{aligned} \Delta unemp_t = & \quad \hat{\alpha}_2 + \hat{\gamma}_{12} \Delta i_t + \hat{w}_t \\ = & \; -0.1 + 0.5 \Delta i_t + \hat{w}_t \\ = & \qquad (1.1) \quad (0.1) \end{aligned} \tag{1}$$

Moreover, he had estimated a regression of your firm's portfolio return on both the change in the fed funds rate and the unemployment rate:

$$\begin{aligned} r_t = & \quad \hat{\alpha} + \hat{\beta}_1 \Delta i_t + \hat{\beta}_2 \Delta unemp_t + \hat{\nu}_t \\ = & \; 6.0 - 1.0 \Delta i_t - 2 \Delta unemp_t + \hat{\nu}_t \\ = & \qquad (5.0)(0.4) \quad (0.5) \end{aligned} \tag{2}$$

where again numbers in parentheses denote the standard errors associated with the estimated coefficients. You have 13 years of data.

<u>Question:</u> Test the hypothesis that both variables, the change in the fed funds rate and the change in the unemployment rate, do not matter for your firm's portfolio at the 5%-level.

What is the null hypothesis in this case? What is the restricted model? If the variance of returns is 25 and you consider 13 years of data, what is the restricted residual sum of squares (RSS)? Further you are given that the RSS of the unrestricted model is 150.

<u>Answer:</u> The null hypothesis is H_0: $\beta_1 = \beta_2 = 0$.
The unrestricted model is $r_t = \alpha + \beta_1 \Delta i_t + \beta_2 \Delta unemp_t + \nu_t$. Under H_0: $\beta_1 = \beta_2 = 0$. Therefore, the restricted model is $r_t = \alpha + \nu_t$.

To compute the RSS of the restricted model, recall that the sample variance of returns is

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^{T}(r_t - \bar{r})^2}{T-1} \tag{3}$$

which can be rearranged to

$$\sum_{t=1}^{T}(r_t - \bar{r})^2 = (T-1)\cdot\hat{\sigma}^2 \tag{4}$$

Note that for our simple restricted regression on a constant, the residuals are the deviation from the mean $\hat{\nu}_t = r_t - \alpha$

$$\text{RSS} = \sum_{t=1}^{T}(r_t - \bar{r})^2 = (T-1)\cdot\hat{\sigma}^2 = 12\cdot 25 = 300 \tag{5}$$

The test-statistic for the F-Test is

$$\begin{aligned}
\text{test statistic} &= \frac{RRSS - URSS}{URSS} \times \frac{T-n}{m} \\
\text{where URSS} &= \text{RSS from unrestricted regression} \\
\text{RRSS} &= \text{RSS from restricted regression} \\
m &= \text{number of restrictions} \\
T &= \text{number of observations} \\
n = k+1 &= \text{\# of regressors in unr. regression incl. constant} \\
&\quad \text{(the total \# of parameters to be estimated)}
\end{aligned}$$

In this question RRRS = 300, URSS = 150 and T-n = 10 and m = 2. Therefore the test-statistic is

$$\text{test statistic} = \frac{300-150}{150} \times \frac{10}{2} = 5$$

The critical value is

$$\text{crit} = F_{0.05\%,m,T-n} = F_{0.05\%,2,10} = 4.103$$

As test statistic > crit we reject H_0 $\beta_1 = \beta_2 = 0$ at the 5-% level.

3 Unconditional Forecast

Assume you are asked to evaluate the risk-return trade-off for a stock X relative to the market MKT. You run regressions of monthly excess returns for the stock (denoted r_X) onto an intercept and the market return (denoted r_M), as well as a regression of the market return onto an intercept and its own lag, respectively. These are your results:

$$r_{X,t} = \underset{(0.04)}{0.1} + \underset{(0.15)}{1.5}\, r_{M,t} + \hat{u}_t \tag{6}$$

$$r_{M,t} = \underset{(0.02)}{0.1} + \underset{(0.06)}{0.1}\, r_{M,t-1} + \hat{w}_t \tag{7}$$

We also know that: $Var(\hat{u}) = \sum_{t=1}^{T} \hat{u}_t^2 = 49$, $Var(\hat{w}) = \sum_{t=1}^{T} \hat{w}_t^2 = 9$, $\sum_{t=1}^{T} \hat{u}_t \hat{w}_t = 0$, $T = 240$. Assume the model is correctly specified and residuals proxy for true errors.

a) Question: Compute the unconditional expected excess returns for X.

Solution: First find the unconditional expected excess return for the market.

$$\begin{aligned} r_{M,t} &= \underset{(0.02)}{0.1} + \underset{(0.06)}{0.1}\, r_{M,t-1} + \hat{w}_t \\ E[r_M] &= E[0.1 + 0.1 r_{M,t-1} + \hat{w}_t] \end{aligned}$$

Recall that in the long run for a stationary time series $E[r_M, t] = E[r_M, t-1]$, then

$$\begin{aligned} E[r_M] &= 0.1 + 0.1 E[r_M] \\ (1 - 0.1)E[r_M] &= 0.1 \\ E[r_M] &= \frac{0.1}{1 - 0.1} = \frac{1}{9} \end{aligned}$$

Then we can use this result to compute $E[r_X]$:

$$\begin{aligned} r_{X,t} &= \underset{(0.04)}{0.1} + \underset{(0.15)}{1.5}\, r_{M,t} + \hat{u}_t \\ E[r_{X,t}] &= E[0.1 + 1.5 r_{M,t} + \hat{u}_t] \\ E[r_X] &= 0.1 + 1.5 E[r_M] \\ E[r_X] &= 0.1 + 1.5 \cdot \frac{1}{9} = \frac{4}{15} \end{aligned}$$

b) Question: Compute the unconditional variance for the excess return of X.

Solution: First find the unconditional variance for the excess return of the market.

$$
\begin{aligned}
r_{M,t} &= \underset{(0.02)}{0.1} + \underset{(0.06)}{0.1}\, r_{M,t-1} + \hat{w}_t \\
Var(r_{M,t}) &= Var(0.1 + 0.1 r_{M,t-1} + \hat{w}_t) \\
Var(r_{M,t}) &= 0 + 0.1^2 \cdot Var(r_{M,t-1)} + Var(\hat{w}_t) \\
Var(r_{M,t}) &= \frac{1}{100} \cdot Var(r_{M,t-1}) + 9
\end{aligned}
$$

Recall, that again in the long-run of a stationary time series we have $Var(r_{M,t}) = Var(r_{M,t-1})$

$$
\begin{aligned}
Var(r_{M,t}) &= \frac{1}{100} \cdot Var(r_{M,t-1}) + 9 \\
Var(r_M) &= \frac{1}{100} \cdot Var(r_M) + 9 \\
(1 - \frac{1}{100}) Var(r_M) &= 9 \\
Var(r_M) &= \frac{9}{1 - \frac{1}{100}} = \frac{100}{11}
\end{aligned}
$$

Then use this result to obtain the unconditional variance for the excess return of X.

$$
\begin{aligned}
r_{X,t} &= \underset{(0.04)}{0.1} + \underset{(0.15)}{1.5}\, r_{M,t} + \hat{u}_t \\
Var(r_{X,t}) &= Var(0.1 + 1.5 r_{M,t} + \hat{u}_t) \\
Var(r_{X,t}) &= 0 + 1.5^2 \cdot Var(r_{M,t}) + Var(\hat{u}_t) \\
Var(r_{X,t}) &= 2.25 \cdot Var(r_{M,t}) + 49 \\
Var(r_{X,t}) &= 2.25 \cdot \frac{100}{11} + 49 = \frac{764}{11}
\end{aligned}
$$